

SAMBHAV SHRESTHA

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EDUCATION

Stony Brook University - SUNY

August 2025 - June 2027

Master's, Computer Science

Computer Vision, Natural Language Processing, Quantum Computing, Computer Architecture & Assembly, Operating Systems

St. Joseph's College New York

September 2018 - June 2022

Bachelor's, Computer Science and Mathematics

GPA: 3.93

Advanced Algorithms & Programming, Advanced Databases, Multivariable Calculus, Linear Algebra, Probability & Statistics

PROFESSIONAL EXPERIENCE

HCL Technologies

March 2024 - July 2025

ML Infra Engineer at Meta

New York, NY, USA

- Architected and deployed **Sherlock**, a production agentic AI system built with fine-tuned LLaMA and RAG to autonomously resolve ML infrastructure queries, reducing escalation volume and accelerating onboarding
- Engineered monitoring systems and a doctor tool using PyTorch to track production LLM health, setting up real-time alerting for entropy explosions, weight deviations, and throughput degradation
- Delivered end-to-end infrastructure support across the full experimentation and model deployment lifecycle, spanning CUDA kernels, distributed training, and model optimization

Tarifica

June 2023 - February 2024

Software/Data Engineer

New York, NY, USA

- Built 300+ web scrapers using Python and BeautifulSoup and designed Flask-based RESTful ETL pipelines to ingest, transform, and route structured data to downstream PostgreSQL analytics systems with strict latency requirements

Amazon

July 2022 - March 2023

Software Development Engineer

Seattle, WA, USA

- Designed and operated mission-critical microservices on AWS (EC2, Lambda, CloudWatch) powering Amazon Go and Just Walk Out retail stores
- Led migration of a high-traffic service to AWS, reducing costs, latency, and support tickets while improving reliability through automated CI/CD pipelines
- Delivered production features across Java, Kotlin, Python, TypeScript, and Ruby in a Linux environment

Microsoft

June 2021 - July 2021

Data Science Student Research Fellow

New York, NY, USA

- Extended the Financial Times police complaints study using R, uncovering novel race and gender patterns across datasets from NYC, Chicago, and Philadelphia
- Built regression and ML models to surface insights, presenting findings through compelling visualizations to data scientists and engineers at Microsoft Research

RESEARCH & PROJECTS

Sherlock — Agentic AI System at Meta

Python, LLaMA, RAG, PyTorch

- Built a production Agentic AI system fine-tuned on Meta's internal Workplace forum data, combining RAG with agentic reasoning to autonomously resolve ML infrastructure queries for engineers and research scientists

Argument Quality Ranking: LLMs vs. Fine-Tuned RoBERTa

PyTorch, Hugging Face, scikit-learn

- Benchmarked fine-tuned RoBERTa against GPT-5.5, Llama 3, and Mistral on pairwise argument quality ranking; developed RoBERTa v3 with margin ranking loss and test-time pair flipping, achieving 0.657 accuracy matching GPT-5.5 (0.665) at a fraction of cost; released on Hugging Face

Frequency Prior for Autoregressive Image Generation

PyTorch, CUDA, Lightning, wandb

- Researched replacing attention-based transformers with Fourier frequency priors to improve stability, convergence, and visual quality in autoregressive image generation

Raft / Paxos Consensus Simulator

Rust, Python, TOML

- Simulated distributed consensus with configurable node counts, network partitions, and leader election; visualized log replication and fault-tolerance under failure scenarios

SKILLS

Languages: Python, PyTorch, Java, Rust, C/C++, Kotlin, TypeScript, R, SQL, Verilog

ML / AI: Transformers, Computer Vision, LLM, CUDA, RAG, TensorFlow, OpenCV, scikit-learn

Cloud & Tools: AWS, GCP, Docker, Flask, PostgreSQL, Linux/Unix